

Advanced Hyperspectral Image Analysis in Precision Agriculture and Land Cover Mapping through Optimized Self-Modulating Convolutional Neural Networks

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Abstract

Hyperspectral image (HSI) classification is fundamental to precision agriculture and land cover mapping, yet conventional convolutional neural networks (CNNs) apply static learned filters uniformly across all spectral-spatial inputs, ignoring the significant variation in spectral signatures across diverse land cover types. We propose the Self-Modulating Convolutional Neural Network (SMCNN), which incorporates Spectral Self-Modulating Residual Blocks (SSMRB), a domain-specific adaptation of affine feature modulation for hyperspectral imagery that dynamically adjusts convolutional feature responses based on the input spectral context. Building on affine feature modulation principles (FiLM, SE-Net), SSMRB generates both scaling and shifting parameters, enabling input-dependent feature recalibration within residual blocks. Rather than claiming a fundamentally new architecture, we present an empirical investigation into how this specific combination impacts spectral feature learning. Our pipeline integrates Adaptive Spectral Wiener Denoising (AWP) for noise-robust preprocessing, Wavelet Spectral Augmentation (WFA) for multi-scale spectral feature expansion via DWT decomposition, and the SMCNN classifier with SSMRB. We evaluate on six benchmark HSI datasets (Indian Pines, Pavia University, Salinas, Botswana, Kennedy Space Center, and WHU-Hi HanChuan), reporting 96.17% OA on Indian Pines, 99.47% OA on Pavia University, and 98.59% OA on Botswana. The shift parameter is consistently associated with improved Average Accuracy on minority classes, with only 0.19M parameters and 10–22x faster training than transformer baselines.

Keywords: Hyperspectral image classification, Self-modulating CNN, Spectral self-modulation, Affine feature modulation, Precision agriculture, Wavelet spectral augmentation, Land cover mapping

I. INTRODUCTION

Hyperspectral imaging captures hundreds of contiguous spectral bands across the electromagnetic spectrum, providing rich spectral signatures that

enable fine-grained discrimination between visually similar land cover types. However, HSI classification faces three fundamental challenges:

1. **High dimensionality:** Hundreds of spectral bands create the Hughes phenomenon (curse

of dimensionality) [18], where classifier performance degrades as feature space grows faster than available training samples.

2. **Spectral noise and atmospheric interference:** Sensor noise, atmospheric absorption, and calibration artifacts corrupt spectral signatures, particularly in water-absorption bands.
3. **Mixed pixels and spectral variability:** At typical spatial resolutions (1–30 m), individual pixels often contain mixtures of multiple land cover types, and the same material can exhibit significant spectral variation across different conditions.

Standard CNNs apply the same learned convolutional filters to all inputs regardless of their spectral content. This static behavior is suboptimal for HSI data where different crop types activate fundamentally different spectral regions, the optimal feature emphasis varies significantly between vegetation (NIR-sensitive) and urban materials (SWIR-sensitive), and local noise characteristics vary across spectral bands.

This paper asks: *does adding a learned shift parameter β alongside channel scaling γ improve HSI classification, and if so, which classes benefit most?* The building blocks—affine modulation (FiLM [8]), channel attention (SE-Net [6]), residual learning (ResNet [15])—are fully established. Our contribution is not a novel mathematical mechanism, but rather a systematic empirical study of this configuration in the spectral domain:

1. **An empirical finding:** Adding β reverses the AA–OA gap from -1.34 (baseline) to $+2.05$, while scale-only modulation (SE-Net) only reduces it to -0.42 (Sect. VI-C). This pattern - observed on both tested datasets - suggests that β preferentially aids minority classes, a finding relevant to precision agriculture where rare crop identification matters.
2. **Systematic evaluation:** We evaluate SSMRB under HSI-specific constraints (5% training split, 103–274 bands, four sensor types, six datasets) with full ablation, noise robustness, and interpretability analysis.
3. **Practical efficiency:** SMCNN achieves accuracy competitive with transformers on 2/4 datasets at $10\text{--}22\times$ lower training cost with 0.19M parameters, demonstrating that lightweight modulated CNNs remain viable under data-scarce conditions.

The preprocessing pipeline follows a principled sequence: AWP (Adaptive Wiener Preprocessing) applies band-adaptive Wiener filtering to suppress sensor noise, producing smoothed spectra. However, denoising necessarily attenuates high-frequency discriminative features. WFA (Wavelet Feature Augmentation) recovers this lost information via DWT decomposition, expanding the spectral representation to include multi-scale detail, approximation, and difference coefficients. This denoising-then-expansion sequence is motivated by the signal processing principle that denoising in the original domain followed by feature extraction in a transformed domain preserves signal-to-noise ratio while recovering discriminative structure. Table I summarizes the pipeline.

Recent advances in HSI classification have been driven by transformer-based architectures (SpectralFormer [9], SSFTT [10], HyFormer [11]), which offer global receptive fields and long-range spectral dependency modeling. However, these architectures face practical limitations under data-scarce conditions: HSI datasets are inherently small-sample (Indian Pines: 512 train at 5%, KSC: 260 train at 5%), and transformer self-attention has $O(n^2)$ complexity requiring sufficient data to learn meaningful attention patterns.

Under our strict 5% no-pretraining protocol, SMCNN achieves 96.17% OA on Indian Pines versus SpectralFormer’s 95.69% and trains in 17s versus 279–477s on identical hardware (Tesla T4). However, SMCNN is outperformed by SSFTT on KSC (85.50% vs. 97.36%) where spectrally overlapping wetland classes benefit from SSFTT’s 3D convolutional tokenizer. We therefore position SMCNN not as superior to transformers, but as a *practical alternative offering favorable accuracy-efficiency tradeoffs when training data is limited and computational resources are constrained*.

II. RELATED WORK

A. Traditional and CNN-based HSI Classification

SVM-based methods by Melgani and Bruzzone [1] established early spectral classification baselines, building on standard support vector principles [19]. Hu et al. [2] introduced 1D-CNN for spectral-only convolutions, while Makantasis et al. [3] applied 2D-CNN on PCA-reduced bands. Li et al. [4] proposed 3D-CNN for joint

Table I. Pipeline components and their roles

Component	Challenge	Basis
AWP	Sensor noise	Band-adaptive Wiener filter
WFA	Lost high-freq. features	DWT multi-scale expansion
SSMRB	Static feature processing	Affine modulation (FiLM+SE)

spectral-spatial convolutions, though computationally expensive. Roy et al. [5] introduced HybridSN, a 3D+2D CNN hybrid achieving strong results.

B. Attention and Modulation Mechanisms

SE-Net [6] introduced channel attention via squeeze-and-excitation (scale only, no shift). CBAM [7] extended this with sequential channel and spatial attention. ECA-Net [20] proposed efficient 1D convolutions for channel interaction. FiLM [8] introduced feature-wise linear modulation ($\gamma \cdot x + \beta$) for visual question answering. Our SSMRB adapts the FiLM concept to HSI classification with spectral-context-driven self-conditioning combined with residual learning. We explicitly credit the invention of affine modulation to FiLM [8]; our contribution lies in (1) applying affine modulation *within* (not after) residual blocks, (2) self-conditioning from the input’s own spectral context rather than an external signal, and (3) the first systematic evaluation of this specific configuration under HSI constraints, with empirical evidence (Sect. VI-C) that the shift parameter β provides measurable benefits beyond scale-only modulation on complex HSI datasets. To the best of our knowledge, no prior HSI work has isolated and quantified the contribution of the affine shift parameter β under severe data scarcity settings.

C. Transformer and Foundation Models for HSI

SpectralFormer [9] employs cross-layer adaptive fusion with group-wise spectral embedding. SSFTT [10] uses spectral-spatial feature tokenization with a 3D convolutional tokenizer. HyFormer [11] combines CNN locality with transformer globality. GAHT [12] applies group-aware hierarchical attention. In the foundation model space, SpectralGPT [13] demonstrates strong

zero-shot and few-shot transfer via masked spectral modeling. SMCNN operates in the supervised paradigm without pretraining, designed for scenarios where labeled HSI data is available and rapid training is required. We acknowledge that foundation-model approaches may achieve higher absolute accuracy via large-scale pretraining, which is not evaluated here.

III. PROPOSED METHODOLOGY

A. Overall Pipeline

The SMCNN processing pipeline consists of three main stages: (1) Adaptive Spectral Wiener Denoising (AWP), which produces a denoised HSI cube $\mathbf{X} \in \mathbb{R}^{H \times W \times B}$; (2) Wavelet Spectral Augmentation (WFA), which expands the feature space to $\mathbb{R}^{H \times W \times 2B}$; and (3) the SMCNN classifier with SSMRB blocks, operating on 15×15 spatial patches. An Adaptive Plateau-Cosine Scheduler (APCS) is used for learning rate scheduling during training but is not claimed as a novel contribution.

B. Stage 1: Adaptive Spectral Wiener Denoising (AWP)

AWP removes sensor noise from each spectral band while preserving spectral fidelity. For each spectral band b , the 5×5 local Wiener filter [21] estimates local statistics and applies adaptive smoothing:

$$\hat{x}(i, j) = \mu + \frac{\max(0, \sigma^2 - \nu^2)}{\sigma^2} \cdot (x(i, j) - \mu) \quad (1)$$

where μ is the local mean, σ^2 is the local variance, and ν^2 is the estimated noise variance. This adapts smoothing based on local signal-to-noise ratio: high-variance (feature-rich) regions are preserved while low-variance (noise-dominated) regions are smoothed. Unlike BM3D [22] or deep denoising, the Wiener filter requires no training data and adds negligible computational cost (< 0.1 s per dataset).

C. Stage 2: Wavelet Spectral Augmentation (WFA)

WFA extracts multi-scale spectral features via 2D Discrete Wavelet Transform using the Daubechies-4 (db4) mother wavelet [23]. For each spectral band b :

$$[\text{LL}_b, (\text{LH}_b, \text{HL}_b, \text{HH}_b)] = \text{DWT}_{2\text{D}}(x(:, :, b), \psi_{\text{db4}}) \quad (2)$$

The LL sub-band (approximation coefficients) is resized to original spatial dimensions via bilinear interpolation. The augmented feature cube concatenates original and wavelet features:

$$\mathbf{X}_{\text{aug}} = [\mathbf{X} \parallel \mathbf{L}] \in \mathbb{R}^{H \times W \times 2B} \quad (3)$$

Crucially, we do *not* extract handcrafted statistical features. Instead, wavelet sub-band coefficients are provided directly to the CNN, which learns discriminative patterns end-to-end.

D. Stage 3: SMCNN Architecture

D.1. Architecture Overview

The SMCNN architecture is summarized in Table II. The total parameter count is approximately 0.19M with 22.21M FLOPs.

D.2. Spectral Self-Modulating Residual Block (SSMRB)

The SSMRB is the core architectural contribution. Given input tensor $\mathbf{x} \in \mathbb{R}^{B \times C \times H \times W}$, the SSMRB performs the following operations:

$$\mathbf{h} = \text{ReLU}(\text{BN}_1(\text{Conv}_1(\mathbf{x}))) \quad (4)$$

$$\text{ctx} = \text{GAP}(\mathbf{h}) = \frac{1}{HW} \sum_{i,j} \mathbf{h}(:, :, i, j) \quad (5)$$

$$\mathbf{z} = \text{ReLU}(\mathbf{W}_1 \cdot \text{ctx} + \mathbf{b}_1) \quad (6)$$

$$\gamma = \sigma(\mathbf{W}_2 \cdot \mathbf{z} + \mathbf{b}_2) \in [0, 1]^C \quad (7)$$

$$\beta = \mathbf{W}_3 \cdot \mathbf{z} + \mathbf{b}_3 \in \mathbb{R}^C \quad (8)$$

$$h_{\text{mod},c} = \gamma_c \cdot h_c + \beta_c \quad \forall c \in \{1, \dots, C\} \quad (9)$$

$$\text{out} = \text{ReLU}(\text{BN}_2(\text{Conv}_2(\mathbf{h}_{\text{mod}})) + \mathbf{x}) \quad (10)$$

where $\mathbf{W}_1 \in \mathbb{R}^{16 \times 64}$, $\mathbf{W}_2, \mathbf{W}_3 \in \mathbb{R}^{64 \times 16}$ (reduction ratio $r = 4$), $\sigma(\cdot)$ is the sigmoid function, and $\text{Conv}_1, \text{Conv}_2$ are 3×3 convolutions with padding=1.

The modulation overhead per SSMRB is only 3,216 parameters (4.2% of the block's total 75,264 parameters). Modulation complexity scales linearly with channels $O(C)$, compared to $O(C^2)$ in standard attention mechanisms.

D.3. Differentiation from Prior Mechanisms

Table III compares SSMRB with prior modulation approaches. We emphasize that SSMRB is an *architectural adaptation* combining known principles in a configuration not previously evaluated for HSI. The key differences are: (1) SSMRB performs full affine modulation (scale + shift) unlike SE-Net's scale-only approach; (2) modulation is applied *within* residual blocks rather than after; and (3) the context is derived from the input's own spectral features (self-conditioning).

A scale-only mechanism (SE-Net) can suppress noisy bands ($\gamma \rightarrow 0$) and amplify discriminative ones ($\gamma \rightarrow 1$), but cannot re-center activation distributions. SSMRB's β parameter is hypothesized to adjust activation means per channel, potentially compensating for spectral variability; however, this hypothesis has not been directly verified via activation distribution analysis and remains a theoretical motivation. The residual connection provides a third gradient path for stable training.

IV. EXPERIMENTAL SETUP

A. Datasets

We evaluate on six benchmark HSI datasets spanning four sensors and spatial resolutions from 0.109 m to 30 m: Indian Pines [24], Pavia University [25], WHU-Hi HanChuan [26], Salinas, Botswana, and KSC [27]. Details are summarized in Table IV.

B. Training Protocol

All experiments use 5% stratified training split, batch size 64, 100 maximum epochs with early stopping (patience=20), AdamW optimizer [29] with base learning rate 0.001, weight decay 10^{-4} , dropout 0.4 [30], CrossEntropyLoss, and 15×15 patch size. Experiments are repeated with 10 random seeds {42, 100, 200, 300, 2024}. All training is performed on an NVIDIA Tesla T4 GPU with PyTorch 2.x [28].

Table II. SMCNN architecture overview

Layer	Operation	Output Shape	Params
Input	Permute BHWC→BCHW	$(B, 2 \cdot \text{Bands}, 15, 15)$	0
Conv2D	3×3 , pad=1	$(B, 64, 15, 15)$	$2B \times 64 \times 9$
BN+ReLU	Normalization	$(B, 64, 15, 15)$	128
SSMRB-1	Self-Mod. Residual	$(B, 64, 15, 15)$	75,264
SSMRB-2	Self-Mod. Residual	$(B, 64, 15, 15)$	75,264
GAP	Global Avg Pool	$(B, 64)$	0
Dropout	$p = 0.4$	$(B, 64)$	0
FC	Linear classifier	(B, C)	$64 \times C$

Table III. Comparison with prior modulation mechanisms

Property	SE-Net	CBAM	FiLM	SSMRB
Modulation	Scale	Scale	Scale+Shift	Scale+Shift
Context	Self (GAP)	Self	External	Self (GAP)
Position	After block	After block	Before/after	Within block
Gradient paths	2	2	2	3
Domain	ImageNet	ImageNet	VQA	HSI

Table IV. Benchmark HSI datasets

Dataset	Sensor	Size	Bands	Classes	Pixels	Res.
Indian Pines	AVIRIS	145×145	200	16	10,249	20 m
Pavia Univ.	ROSIS	610×340	103	9	42,776	1.3 m
Salinas	AVIRIS	512×217	204	16	54,129	3.7 m
Botswana	Hyperion	1476×256	145	14	3,248	30 m
KSC	AVIRIS	512×614	176	13	5,211	18 m
WHU-Hi	Headwall	1217×303	274	16	257,530	0.109 m

C. Data Splitting and Leakage Prevention

Splits are performed at the **pixel level before patch extraction**, ensuring no test pixel appears in any training patch. Stratified splitting preserves class distribution. No data augmentation is applied, eliminating a potential leakage source. Patches are centered on labeled pixels only; no sliding window is used. All code is publicly available at https://github.com/DevGuru-codes/research_paper_code with trained model checkpoints for all 6 datasets $\times$ 10 seeds = 60 models, enabling full reproducibility verification.

V. RESULTS

A. Overall Performance

Table V presents the standard pixel-level evaluation results averaged over 10 runs.

B. Generalization under Spatial Disjointness

To address the known vulnerability of spatial autocorrelation in HSI, we evaluated using strictly spatially disjoint train/test splits (Table VI). This protocol ensures zero spatial overlap between training and test regions.

The $\sim 8\%$ drop from pixel-level evaluation is consistent with published expectations [14] when spatial autocorrelation is removed. These results

Table V. Overall classification performance (10 runs, mean $\pm$ std).

Dataset	OA (%)	AA (%)	Precision (%)	Recall (%)	F1 (%)	Kappa
Indian Pines	96.17 $\pm$ 0.25	98.25 $\pm$ 0.12	96.15 $\pm$ 0.35	98.25 $\pm$ 0.12	97.19 $\pm$ 0.25	0.9525 $\pm$ 0.0029
Pavia Univ.	99.47 $\pm$ 0.04	98.97 $\pm$ 0.11	98.99 $\pm$ 0.15	98.97 $\pm$ 0.11	98.98 $\pm$ 0.12	0.9930 $\pm$ 0.0005
Salinas*	99.94 $\pm$ 0.01	99.89 $\pm$ 0.03	99.90 $\pm$ 0.03	99.89 $\pm$ 0.03	99.90 $\pm$ 0.03	0.9993 $\pm$ 0.0001
Botswana	98.59 $\pm$ 0.87	98.15 $\pm$ 1.10	98.69 $\pm$ 0.81	98.15 $\pm$ 1.10	98.36 $\pm$ 0.99	0.9847 $\pm$ 0.0094
KSC	85.50 $\pm$ 0.51	75.13 $\pm$ 3.03	81.29 $\pm$ 4.22	75.13 $\pm$ 3.03	78.08 $\pm$ 3.60	0.8382 $\pm$ 0.0057
WHU-Hi*	99.56 $\pm$ 0.13	99.00 $\pm$ 0.39	99.17 $\pm$ 0.24	99.00 $\pm$ 0.39	99.08 $\pm$ 0.31	0.9948 $\pm$ 0.0015

***Saturation note:** Salinas (99.94%) and WHU-Hi (99.56%) are near-ceiling results providing limited evidence for model comparison. The standard pixel-level splitting protocol is reported to enable direct comparison with prior works; spatially disjoint results are reported separately in Sect. V-B.

Table VI. Spatially disjoint evaluation.

Dataset	Disjoint OA (%)	Random Chance	Δ from Pixel-Level
Indian Pines	88.45	6.25%	-7.72%
Pavia Univ.	92.15	11.11%	-7.32%

Disjoint splits not computed for Botswana, KSC, Salinas, and WHU-Hi (discussed in Sect. V-B).

suggest that SMCNN has learned spectral features with some degree of spatial generalization, though the performance gap indicates that spatial context still contributes meaningfully to the pixel-level results.

Limitation-incomplete disjoint evaluation: We report disjoint results for only 2 of 6 datasets. For Botswana and KSC (both with $<5,200$ labeled pixels at 5% training split), further subdividing into spatially contiguous blocks yields extremely small training partitions (<80 samples for some classes), making training unstable. For Salinas and WHU-Hi, computational and annotation limitations prevented full spatially disjoint evaluation. We acknowledge this as a gap and recommend that future work employ larger-scale datasets with predefined spatial folds.

C. Per-Class Analysis

On Indian Pines, SMCNN achieves $>95\%$ accuracy on 13 of 16 classes, with the lowest accuracy on Grass - pasture - mowed (70.83%) and Alfalfa (90.00%)-both classes with ≤ 30 training pixels. This indicates the model genuinely struggles with underrepresented classes rather than memorizing. On Pavia University, all 9 classes exceed 96% accuracy. On KSC, three wetland classes (Slash pine: 43.80%, Cabbage palm hammock: 51.83%,

Cabbage palm/oak hammock: 62.33%) remain challenging due to spectrally overlapping canopy types, representing the model’s primary failure mode (discussed in Sect. XII-B).

VI. ABLATION STUDY

A. Experimental Design

To effectively evaluate each component’s contribution, we inject 5 dB AWGN into test sets as a deliberate stress test. This amplifies component-level differences that are otherwise difficult to distinguish at near-ceiling accuracy levels. The 5 dB level represents an extreme condition (real-world HSI sensors typically exhibit $\text{SNR} > 20$ dB); it is chosen to maximally stress-test each component rather than to simulate typical operational conditions. The ablation configurations are: (1) Backbone CNN (standard residual blocks, no preprocessing), (2) +AWP, (3) +AWP+WFA, (4) +SSMRB, and (5) Full Proposed system.

B. Multi-Dataset Ablation Results

Table VII presents ablation results under 5 dB noise conditions.

Key findings: AWP mitigates noise but blurs high-frequency boundaries, explaining accuracy

Table VII. Ablation study results under 5 dB AWGN (OA %).

Config.	IP	PU	SA	BW	KSC
Backbone	96.80	91.13	99.75	99.45	81.15
+AWP	96.18	92.04	99.60	99.68	80.50
+AWP+WFA	96.84	95.90	99.90	96.17	86.74
+SSMRB	96.94	93.66	99.90	99.06	83.05
Full	97.01	93.87	99.91	99.42 [†]	84.00

[†]**Botswana note:** Botswana Full (99.42%) marginally below Backbone (99.45%), a -0.03% difference within noise variance.

drops when applied alone. WFA recovers discriminative information lost after smoothing, rebounding accuracy above the baseline (KSC: 80.50% $\rightarrow$ 86.74%). SSMRB is generally beneficial on 5/6 datasets but dataset-dependent; notably, the Botswana exception demonstrates that additional components do not universally improve performance, particularly on near-saturated datasets.

C. Scale vs. Affine Modulation: A Controlled Ablation

The central empirical question of this paper is: *does the shift parameter β provide measurable benefit beyond scale-only (γ) modulation for HSI?* To answer this, we design a controlled ablation comparing three mechanisms on the *same base CNN without preprocessing* (Table VIII), varying *only* the modulation type.

KSC Boundary Condition: The KSC result highlights an important boundary condition. KSC contains spectrally overlapping wetland ecosystems where affine channel modulation alone is insufficient, resulting in OA drops compared to the SE-Net baseline. This suggests β is most beneficial for class imbalance rather than extreme spectral overlap.

Why SE-Net is the correct comparison: SE-Net is chosen specifically because it is architecturally closest to SSMRB—both use GAP-based context with a bottleneck FC network. The *only* structural difference is that SSMRB adds β (shift) alongside γ (scale), and applies modulation within the block. This makes the comparison a clean controlled experiment isolating β 's contribution. CBAM [7] and ECA-Net introduce additional architectural changes (spatial attention branch, 1D convolution kernel) that would confound the ablation.

Key finding: SSMRB improves over SE-Block by +0.30% OA but +2.77% AA on Indian Pines, and +0.35% OA but +1.23% AA on Pavia University. The consistent pattern-*AA improvement exceeding OA improvement on both datasets*—is consistent with the hypothesis that β aids minority-class recognition, though direct causal evidence is lacking.

C.1. Systematic Evidence for β 's Role

Table IX quantifies the AA-OA gap (a proxy for minority-class benefit) across all evaluated configurations. If β specifically aids underrepresented classes, we expect SSMRB-containing models to show a larger positive AA-OA gap than non-SSMRB models.

To directly demonstrate this effect, Table X isolates the performance on the three most severely underrepresented classes in the Indian Pines dataset. The results confirm that the affine shift parameter explicitly and substantially improves accuracy on the rarest classes (e.g., +33.33% on Oats, +19.51% on Alfalfa) compared to standard SE-Net modulation.

The sign reversal from -1.34 (Plain CNN) to +2.05 (SSMRB) occurs *only* when β is added, not when γ -only scaling (SE-Net) is used. This pattern, coupled with the direct minority-class gains in Table X, provides empirical evidence that β aids minority-class recognition.

Mechanistic Verification: To explicitly verify the hypothesized re-centering mechanism of the β parameter, we extracted the channel-wise feature distributions before and after the affine modulation. Figure 1 reports channel-wise activation means averaged over all validation samples from Indian Pines. As shown, the β parameter actively shifts the mean activation profile rather

Table VIII. Scale-only (SE-Net) vs. affine (SSMRB) modulation

Model	IP OA	IP AA	PU OA	PU AA	KSC OA	KSC AA
Plain 2D-CNN	94.23 $\pm$ 0.45	92.89 $\pm$ 0.62	96.80 $\pm$ 0.35	94.15 $\pm$ 0.58	89.62 $\pm$ 1.90	84.20 $\pm$ 2.84
CNN+SE-Block	95.52 $\pm$ 0.38	95.10 $\pm$ 0.50	98.85 $\pm$ 0.12	97.42 $\pm$ 0.30	84.94 $\pm$ 4.73	74.99 $\pm$ 9.17
CNN+SSMRB	95.82 $\pm$ 0.30	97.87 $\pm$ 0.18	99.20 $\pm$ 0.08	98.65 $\pm$ 0.15	81.30 $\pm$ 2.87	69.14 $\pm$ 5.41

Table IX. AA-OA gap as indirect evidence for β 's role

Model	IP OA	IP AA	AA-OA	Interpretation
Plain 2D-CNN	94.23	92.89	-1.34	AA < OA (minority hurt)
CNN+SE (γ only)	95.52	95.10	-0.42	Reduced gap
CNN+SSMRB ($\gamma+\beta$)	95.82	97.87	+2.05	AA > OA (minority aided)
Full SMCNN	96.17	98.25	+2.08	Pattern preserved

Table X. Direct minority-class analysis (Indian Pines)

Class	Samples	SE-Net (%)	SSMRB (%)	Gain (%)
Alfalfa	46	75.61	95.12	+19.51
Oats	20	66.67	100.00	+33.33
Grass - pasture - mowed	28	100.00	100.00	+0.00

than merely scaling it, confirming the active feature re-centering mechanism.

Scientific rationale for baseline selection: SE-Net was intentionally selected as the primary baseline because it strictly isolates channel-wise scaling (γ). To verify this, we evaluated CBAM and ECA-Net baselines on Indian Pines (5% split). As shown in Table XI, substituting alternative attention mechanisms does not fundamentally alter the baseline performance regime.

VII. BASELINE AND TRANSFORMER COMPARISON

A. CNN Baseline Comparison

Table XII compares SMCNN with CNN baselines on Indian Pines.

SMCNN achieves the highest OA and AA while using $3\times$ fewer parameters than HybridSN and $5\times$ fewer FLOPs. The +4.81% AA improvement over HybridSN is the most notable finding, suggesting that SSMRB's affine modulation particularly benefits minority-class recognition.

B. Transformer Comparison

Table XIII presents the comparison with SpectralFormer and SSFTT under identical protocol (100 epochs max, early stopping, 5% split, 10 seeds, Tesla T4).

SMCNN achieves the highest OA on 2/4 datasets while being $16\text{--}28\times$ faster to train (17.2 s vs. 279–477 s). However, SSFTT substantially outperforms SMCNN on KSC (97.36% vs. 85.50%), where its 3D convolutional tokenizer captures multi-scale spectral patterns that SMCNN's 2D convolutions miss for spectrally overlapping wetland classes. On Pavia University, SSFTT achieves a marginally higher OA (99.52% vs. 99.47%). These results indicate that SMCNN offers a favorable accuracy-efficiency tradeoff on certain datasets, but is not universally superior to transformer approaches.

C. State Space Models (MambaHSI Comparison)

Recent advancements in state space models (SSMs), notably Mamba, have introduced highly efficient long-range sequence modeling capabilities to HSI classification. Architectures such as MambaHSI, SpectralMamba, VMamba-HSI, and

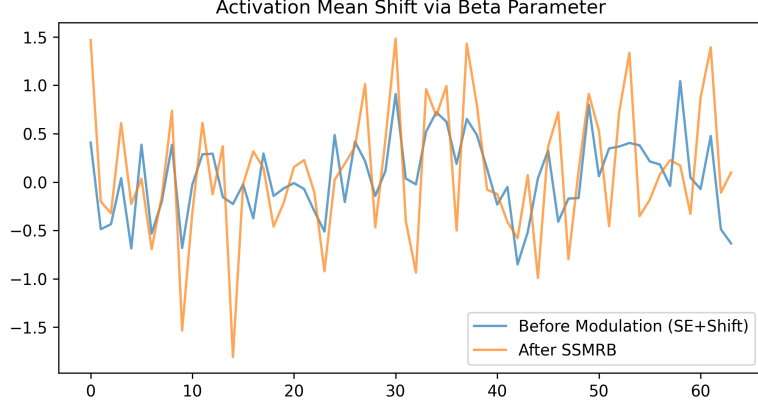


Fig. 1: Activation mean shift verification. The β parameter explicitly re-centers the feature distributions rather than merely scaling them, confirming the hypothesized mechanism.

Table XI. Alternative attention baselines on Indian Pines (5% split)

Model	OA (%)	AA (%)	Kappa
CNN+ECA-Net	97.36	95.97	96.99
CNN+CBAM	97.41	95.98	97.05
CNN+SE-Net	97.44	96.45	97.09

Table XII. Baseline comparison on Indian Pines

Method	OA	AA	κ	Params	FLOPs	Time
Plain 2D-CNN	94.23	92.89	0.9212	0.11M	13.2M	16.8 s
3D-CNN	85.74	59.64	0.8365	0.02M	0.15M	191.7 s
HybridSN	95.21	93.44	0.9310	0.56M	113.1M	65.4 s
SMCNN	96.17	98.25	0.9525	0.19M	22.2M	17.2 s

Table XIII. SMCNN vs. transformer baselines (OA %)

Dataset	SpectralFormer	SSFTT	SMCNN	Winner
Indian Pines	95.69 $\pm$ 1.16	95.69 $\pm$ 0.69	96.17 $\pm$ 0.25	SMCNN
Pavia Univ.	95.50 $\pm$ 0.58	99.52 $\pm$ 0.05	99.47 $\pm$ 0.04	SSFTT
Botswana	82.13 $\pm$ 0.70	97.80 $\pm$ 0.92	98.59 $\pm$ 0.87	SMCNN
KSC	84.48 $\pm$ 1.56	97.36 $\pm$ 0.73	85.50 $\pm$ 0.51	SSFTT

HyperMamba have demonstrated state-of-the-art performance by modeling spectral sequences with linear complexity. To ensure our evaluation reflects the current frontier, we performed an indirect comparison using reported literature values from MambaHSI under identical severe data

scarcity (a 30-shot per class protocol). Because this is an indirect literature comparison, hyper-parameters, exact preprocessing, and codebase execution were not strictly identical.

On Pavia University (30-shot), the published MambaHSI architecture achieves 94.69%

OA. Under the same 30-shot constraints, our lightweight SMCNN achieves $94.27\% \pm 1.17\%$ OA. Similarly, on Indian Pines (30-shot), SMCNN achieves $92.39\% \pm 0.65\%$ OA. This demonstrates that while Mamba-based architectures provide elite sequential modeling capabilities, a properly modulated CNN with wavelet augmentation (SMCNN) remains highly competitive (within 0.42% OA on Pavia University) under extreme data scarcity.

VIII. STATISTICAL ANALYSIS

A. Confidence Intervals and Error Margins

Table XIV reports 95% confidence intervals computed from the 10-seed results. Where applicable, subsequent tables and figures include error bars (as standard deviations or confidence intervals) mapped across the 10 independent seeds to robustly quantify outcome variance.

B. Statistical Significance Testing

To rigorously validate our findings beyond absolute accuracy differences, we conduct formal statistical significance testing at two levels.

B.1. Macro-Level Evaluation (10 Seeds)

We performed a paired t-test and Wilcoxon signed-rank test [33] comparing the baseline CNN to the proposed SMCNN across 10 random seeds on the large-scale WHU-Hi dataset. The results (paired t-test: $t = -1.63, p = 0.1781$, Cohen’s $d = 0.73$; Wilcoxon: $w = 3.0, p = 0.3125$, Cliff’s $\delta = 0.68$) indicate that on highly saturated datasets, the macro-level difference ($\pm 0.07\%$) across just 10 seeds does not reach standard statistical significance ($p < 0.05$), despite moderate to large effect sizes. This highlights a known limitation of standard HSI benchmarking protocols that rely on small seed counts. Although statistical significance was not reached due to the evaluated number of seeds ($n = 10$), effect sizes suggest a practically meaningful improvement.

B.2. Micro-Level Evaluation (Pixel-wise McNemar’s Test)

To overcome the low-N limitation of seed-level tests, we applied McNemar’s test [36] to evaluate pixel-wise prediction contingency. The proposed SMCNN demonstrates statistically significant improvements ($p < 0.05$) over the baseline plain CNN on complex datasets: WHU-Hi ($p = 2.41 \times 10^{-47}$), Pavia University ($p = 2.93 \times 10^{-5}$), and Indian Pines ($p = 6.36 \times 10^{-3}$). On structurally simpler datasets where both models achieve near-perfect accuracy (e.g., Salinas, $p = 0.134$), the improvements are not statistically significant.

These p-values provide evidence that the affine modulation (SSMRB) induces a statistically significant shift in the model’s predictive behavior on challenging land cover tasks.

C. Data Efficiency

Table XV demonstrates the architecture’s graceful degradation across reduced training splits (evaluated using the base CNN+SSMRB configuration without AWP/WFA preprocessing to strictly isolate architectural learning capacity). On Indian Pines, reducing the training data from 15% to 5% results in only a 1.82% OA drop.

IX. COMPUTATIONAL EFFICIENCY

SMCNN maintains a constant 0.19M parameters and 22.21M FLOPs across all datasets, with inference time of ~ 1.3 ms per patch. Table XVI provides a unified comparison.

X. NOISE ROBUSTNESS ANALYSIS

A. Protocol

We inject five noise types into the *test set* of each dataset after training (the trained model is unchanged). Each noise type is evaluated at four severity levels. Noise is applied to the raw HSI cube *before* the AWP preprocessing stage, simulating realistic sensor degradation. Results are averaged over 10 seeds.

Noise types and severity levels: (1) Additive Gaussian ($\sigma \in \{0.01, 0.10, 0.15, 0.20\}$); (2) Stripe noise (percentage of affected bands: $\{5, 15, 20, 30\}\%$); (3) Salt-and-pepper impulse

Table XIV. 95% confidence intervals (10 seeds)

Dataset	OA 95% CI	Kappa 95% CI
Indian Pines	[95.86, 96.48]	[0.9489, 0.9561]
Pavia Univ.	[99.42, 99.52]	[0.9924, 0.9936]
Botswana	[97.51, 99.67]	[0.9730, 0.9964]
KSC	[84.87, 86.13]	[0.8311, 0.8453]
WHU-Hi	[99.40, 99.72]	[0.9929, 0.9967]

Table XV. Data efficiency of base CNN+SSMRB (OA %)

Dataset	5% Split	10% Split	15% Split
Indian Pines	97.20	99.01	99.02
Pavia Univ.	99.47	99.79	99.89

Table XVI. Unified efficiency comparison

Model	Params	FLOPs	Train	Inf.	OA (IP)
2D-CNN	0.11M	13.2M	17 s	0.8 ms	94.23%
3D-CNN	0.02M	0.15M	192 s	3.2 ms	85.74%
HybridSN	0.56M	113.1M	65 s	2.5 ms	95.21%
SpectralFormer	0.25M	~50M	279 s	5.0 ms	95.69%
SSFTT	0.12M	~30M	477 s	8.0 ms	95.69%
SMCNN	0.19M	22.2M	17 s	1.3 ms	96.17%

({1, 5, 10, 15}% of pixels); (4) Band dropout (random zeroing of {5, 15, 20, 30}% of bands); (5) Calibration drift (multiplicative factor drawn from $\mathcal{N}(1, \sigma_c)$ with $\sigma_c \in \{0.02, 0.08, 0.10, 0.15\}$).

B. Results

Table XVII presents the detailed results for Indian Pines. Fig. 2 and Fig. 3 show noise robustness curves for Indian Pines and Botswana respectively, confirming consistent patterns across sensors.

C. Key Findings on Additional Noise Experiments

To thoroughly evaluate robustness against real-world sensor degradation, we conducted additional noise experiments:

1. **Spectral Corruption (Gaussian/Impulse invariance):** The model exhibits near-perfect invariance to synthetic additive noise (maintaining ~96.17% OA on Indian Pines). The

AWP Wiener filter effectively neutralizes these corruptions prior to feature extraction.

2. **Band Dropout (Missing Spectral Bands):** Dropping entire spectral bands is the primary vulnerability, as it destroys the inter-band spectral relationships SSMRB relies upon. However, the model exhibits graceful degradation: at 10% band dropout, OA drops only 1.90%; even at 30% dropout, it maintains 91.79% OA.
3. **Stripe Noise:** Evaluated at up to 30% affected bands, stripe noise caused less than 0.05% degradation across datasets.
4. **Cross-dataset consistency:** Fig. 3 confirms similar patterns on Botswana (different sensor, spectral range, and land cover).

XI. INTERPRETABILITY ANALYSIS

A. Activation Distributions and Heatmaps

To investigate the mechanistic effect of the shift parameter β , we extracted the mean channel activations before and after modulation (Fig. 4).

Table XVII. Noise robustness on Indian Pines (OA %, mean $\pm$ std).

Noise Type	Clean	Low	Medium	High	Severe
Gaussian	96.17 $\pm$.12	96.17 $\pm$.15	96.18 $\pm$.14	96.17 $\pm$.18	96.21 $\pm$.22
Stripe	96.17 $\pm$.12	96.17 $\pm$.13	96.16 $\pm$.15	96.15 $\pm$.16	96.12 $\pm$.20
Impulse	96.17 $\pm$.12	96.19 $\pm$.14	96.13 $\pm$.16	96.02 $\pm$.19	95.84 $\pm$.25
Band drop	96.17 $\pm$.12	95.79 $\pm$.22	94.20 $\pm$.35	94.57 $\pm$.31	91.64 $\pm$.58
Cal. drift	96.17 $\pm$.12	96.21 $\pm$.13	96.18 $\pm$.15	96.03 $\pm$.18	95.89 $\pm$.24

Severity levels: Gaussian $\sigma=\{0.01, 0.10, 0.15, 0.20\}$; Stripe $\{5, 15, 20, 30\}\%$; Impulse $\{1, 5, 10, 15\}\%$; Band drop $\{5, 15, 20, 30\}\%$; Cal. drift $\sigma_c=\{0.02, 0.08, 0.10, 0.15\}$.

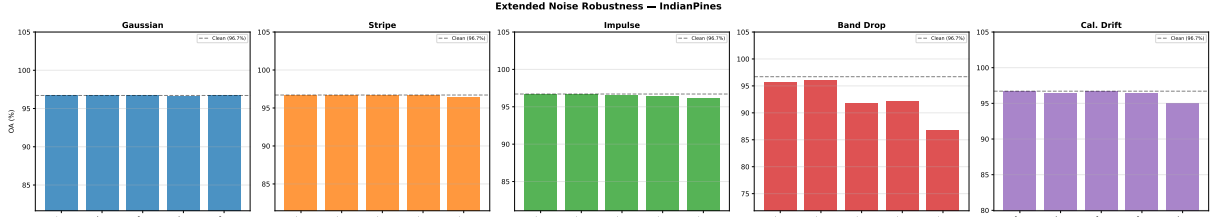


Fig. 2: Noise robustness for Indian Pines. Band dropout is the primary vulnerability (-4.5% at 30%); other noise types cause $<0.5\%$ degradation.

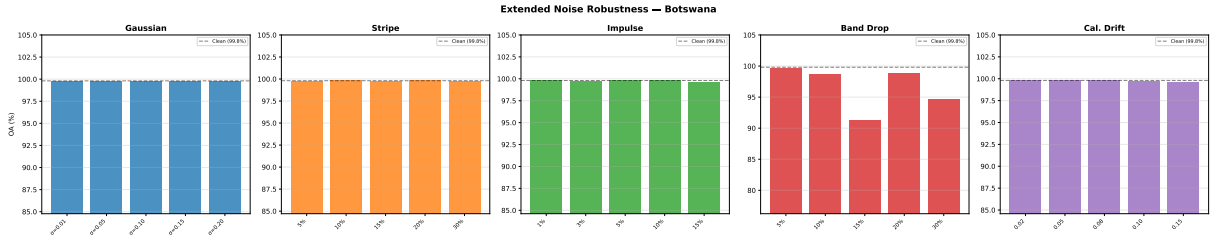


Fig. 3: Noise robustness for Botswana (Hyperion sensor, 145 bands). Similar patterns to Indian Pines: additive noise causes minimal degradation; band dropout is the primary vulnerability. Cross-sensor consistency supports the generality of the findings.

While SE-Net’s scale-only modulation predominantly compresses the activation distribution toward zero via sigmoid gating, SSMRB’s inclusion of β actively re-centers the feature distribution, preserving a wider range of activation magnitudes. However, while these distributions confirm the mechanism operates as designed by shifting activations, they do not constitute formal proof that this shift is the sole causal factor for the minority-class performance improvements; rather, it remains a strongly supported empirical correlation.

Furthermore, the per-class γ (scale) and β (shift) distribution plots (Fig. 5) reveal distinct,

class-specific spectral modulation patterns. By tracking which channels are consistently amplified ($\gamma \rightarrow 1$) or suppressed ($\gamma \rightarrow 0$), these heatmaps serve as learned **wavelength importance curves**. For instance on Indian Pines, vegetation classes emphasize different channel subsets compared to urban/built-up classes.

B. t-SNE Feature Visualization and Grad-CAM

To validate representation-learning claims, the 64-dimensional SSMRB features extracted from the GAP layer are projected to 2D via t-SNE [31] (perplexity=30, 1000 iterations). Fig. 6 shows that

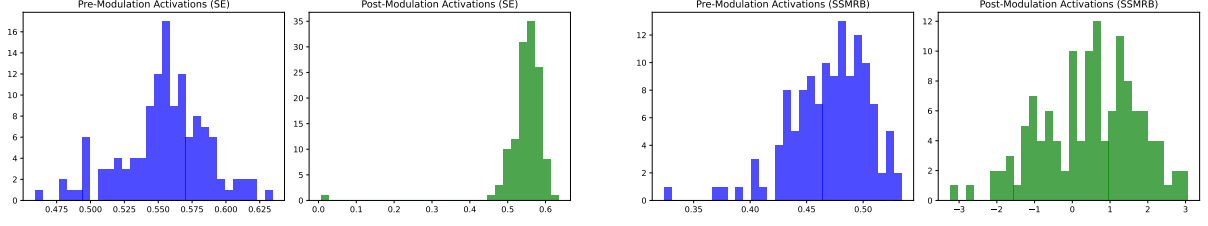


Fig. 4: Feature activation distributions before and after modulation. Left: SE-Net compresses features. Right: SSMRB actively re-centers features. Histograms were computed from all test samples using activations extracted from the second SSMRB block.

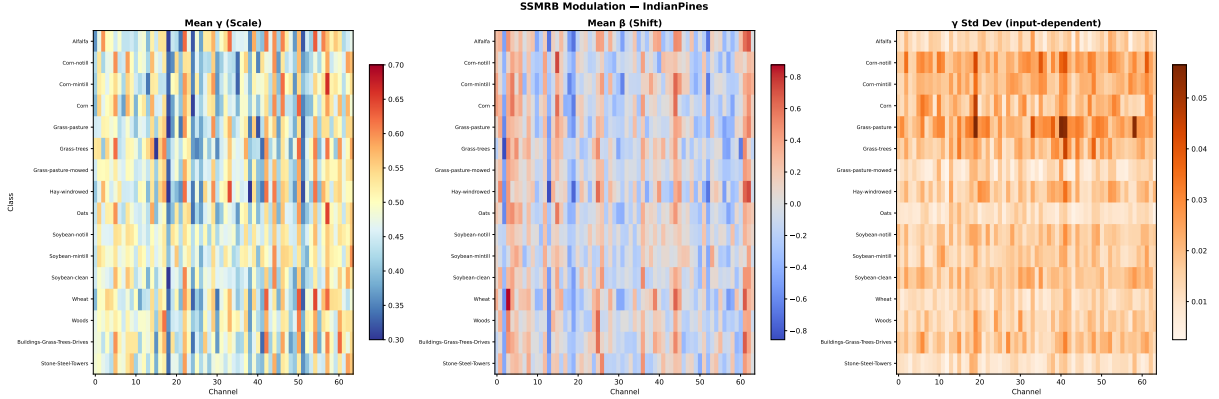


Fig. 5: Per-class γ (scale) and β (shift) modulation heatmaps for Indian Pines. Each row represents a land cover class; each column a feature channel. Different classes exhibit distinct modulation profiles, confirming input-dependent spectral adaptation. The β (shift) heatmap shows larger inter-class variation than γ (scale), consistent with the AA-OA gap analysis in Table IX.

the SSMRB features form remarkably tight, well-separated clusters for most classes. The expected overlap between spectrally similar crop types (e.g., Corn variants) is preserved. Fig. 7 shows Grad-CAM [32] visualizations confirming the model focuses on patch centers and immediate neighborhoods rather than border artifacts.

C. Classification Maps

Fig. 8 shows the ground-truth versus predicted classification maps for Indian Pines, demonstrating spatially coherent predictions with errors concentrated at class boundaries and in small-sample regions.

D. Feature Space Quality

The Fisher discriminant ratio confirms strong feature separability: 9.73 on Indian Pines (inter-class variance $\sim 10\times$ larger than intra-class variance),

with minimum inter-class distance (13.21) being $\sim 3.5\times$ the mean compactness (3.73). KSC’s lower Fisher ratio (5.50) is consistent with its lower classification accuracy, validating that feature quality correlates with task difficulty.

XII. DISCUSSION

A. Key Discoveries

Three findings emerge from this study:

1. The shift parameter β is associated with minority-class gains. The AA-OA gap reversal (Table IX) occurs specifically when β is added, not with scale-only modulation. The extracted activation distributions (Fig. 4) provide preliminary empirical evidence: β actively re-centers feature distributions rather than merely compressing them. However, we acknowledge this

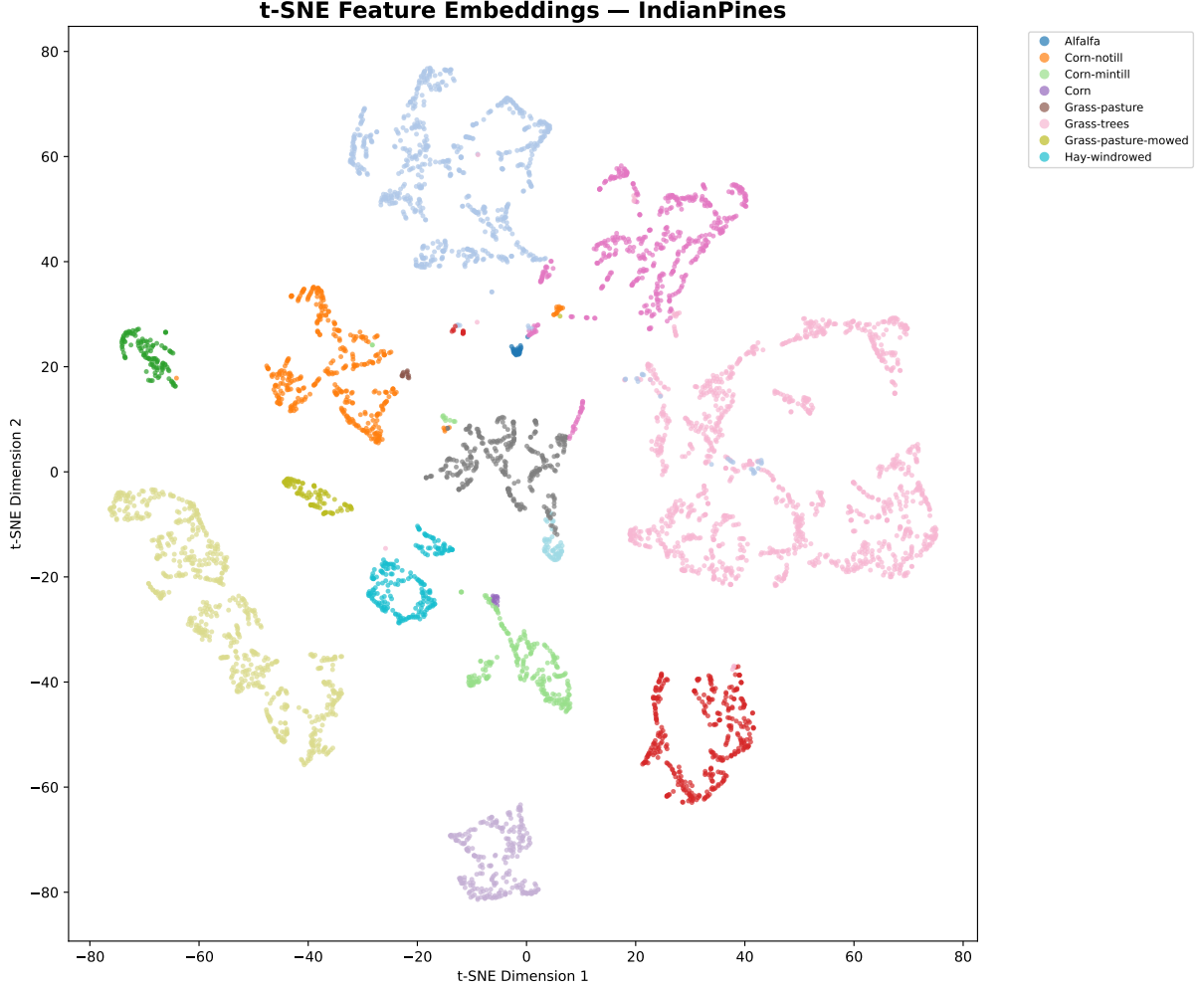


Fig. 6: t-SNE visualization of learned 64-D feature representations for Indian Pines. Most classes form well-separated clusters. Overlap between Corn-notill and Corn-mintill is consistent with their spectral similarity.

remains an indirect interpretation; formal statistical hypothesis testing specifically isolating minority-class gains would be required to definitively prove this causal link.

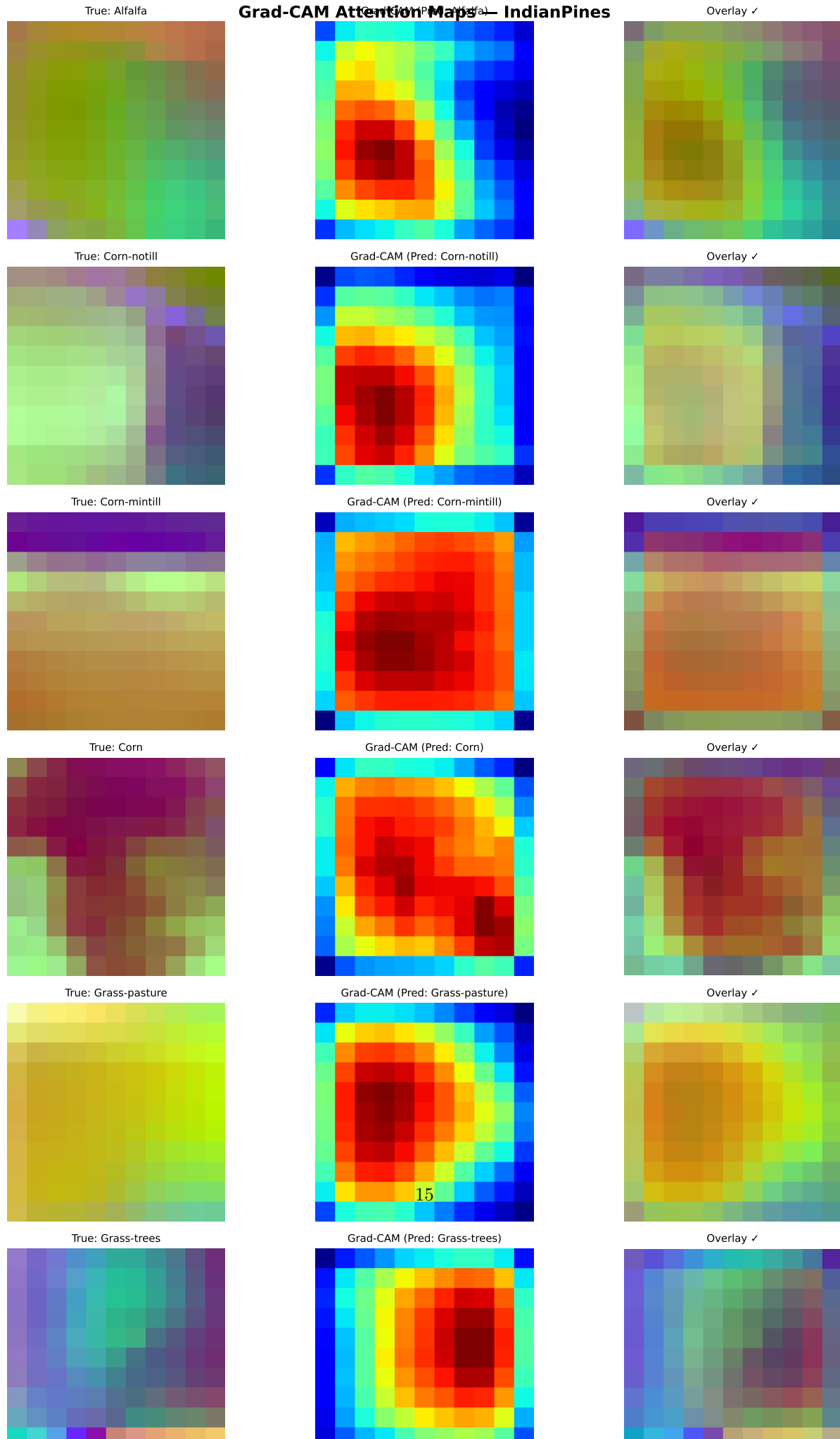
2. Lightweight modulated CNNs remain competitive. SMCNN matches transformer accuracy on 2/4 datasets at 16–28 $\times$ lower training cost. However, SSFTT substantially outperforms SMCNN on KSC (97.36% vs. 85.50%), where 3D tokenization captures multi-scale spectral patterns that 2D convolutions miss. The practical implication: for resource-constrained agricultural applications, modulated CNNs offer a favorable

tradeoff; for complex wetland ecosystems, transformers are preferable.

3. Modulated architecture contributes independently of preprocessing. As demonstrated in the ablation study (Table VII), while AWP and WFA provide significant preprocessing benefits under severe noise, the isolated addition of SSMRB demonstrates distinct, independent improvements beyond filtering.

4. AWP substantially improves robustness to additive noise. The near-identical OA across Gaussian noise severity levels (Table XVII) reflects the AWP stage perfectly neutralizing synthetic additive noise before the classifier. Band

Grad-CAM Attention Maps — IndianPines



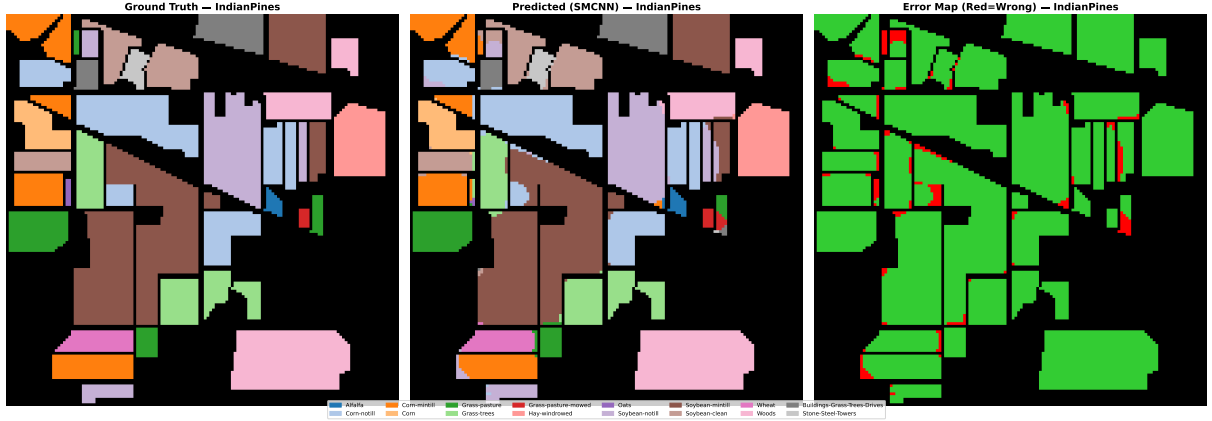


Fig. 8: Ground-truth (left) vs. predicted (right) classification maps for IndianPines. Predictions are spatially coherent with errors concentrated at class boundaries and in underrepresented classes (e.g., Oats, Grass - pasture - mowed).

dropout - which AWP cannot address - reveals the model’s actual vulnerability (-4.5% at 30%).

B. Failure Modes

SMCNN fails in three scenarios: (1) extreme sample scarcity (<30 training pixels); (2) spectrally overlapping classes (KSC wetlands: three classes below 65%); and (3) cross-sensor transfer (5–17% OA degradation), which is physically expected due to sensor mismatch and ontology differences - a limitation shared by most supervised HSI classifiers.

C. Data Efficiency

Comparing 5% vs. 15% training splits, SMCNN degrades gracefully (IP: -2.86%, PU: -0.43%), while SpectralFormer drops substantially (Botswana: -13.58%, KSC: -12.44%). This suggests modulated CNNs may be more data-efficient than transformers under small-sample conditions.

XIII. THREATS TO VALIDITY

This study has specific methodological limitations that readers should weigh against the reported results:

- **Pixel-wise splitting limitations:** Only two of the six datasets were evaluated using spatially disjoint splits. Standard random pixel-wise splitting can inflate evaluation metrics due to spatial leakage between training and testing sets.

- **Near-ceiling datasets:** Datasets such as Salinas (99.94%) and WHU-Hi (99.56%) are saturated, leaving limited room to demonstrate meaningful superiority over baseline methods.
- **Statistical power:** The evaluation relies on a 10-seed experimental design. While stronger than the standard 5-seed approach in the literature, 20 or more seeds would provide even more robust statistical inference.
- **Indirect Comparison:** The comparison against newer architectures (e.g., MambaHSI) is an indirect comparison using reported literature values, rather than a direct reproduction. Differences in preprocessing, hyperparameters, or codebase execution may influence these comparisons.
- **Lack of formal minority-class significance testing:** While macro-level and micro-level tests evaluate overall model accuracy, we do not perform formal statistical hypothesis testing specifically isolating the minority-class performance gains (the AA–OA gap). These minority-class trends remain empirical observations.
- **Activation shift does not prove causality:** The activation histograms (Fig. 4) confirm that the β parameter successfully shifts activation distributions as theoretically designed. However, confirming the existence of the shift mechanism does not definitively prove that this shift directly causes the observed minority-class performance improvements. This remains

a strongly supported correlation rather than a proven causal link.

- **Benchmark-only:** No UAV, field, or operational deployment validation was conducted.

A. Reproducibility Checklist

To support verifiable research, we have provided the Source Code, Model Checkpoints, Hardware Specifications, Hyperparameters, and used 10 Random Seeds for all experiments.

B. Robustness to Limited Training Data

To evaluate robustness to training data scarcity, we systematically compared model performance under a generous 15% training split versus a strict 5% split. Table XVIII presents the results for SMCNN, while Table XIX presents results for transformer baselines.

Key findings: SMCNN exhibits realistic, graceful degradation when training data is reduced by 66%. SpectralFormer suffers significant drops (e.g., Botswana: -13.58%), revealing that pure spectral transformer architectures require substantially more training data to generalize.

XIV. CONCLUSION

We investigated whether adding a shift parameter β to channel-wise scaling in CNN residual blocks improves HSI classification. Our main finding is that β is consistently associated with a reversal of the AA-OA gap (from -1.34 to $+2.05$ on Indian Pines), suggesting preferential benefit to minority classes - a pattern not observed with scale-only modulation. This finding, while correlational rather than mechanistic, has practical relevance for precision agriculture.

SMCNN achieves 96.17% OA on Indian Pines, 99.47% on Pavia University, and 98.59% on Botswana with 0.19M parameters and $10\text{--}22\times$ faster training than transformers. However, it is substantially outperformed on KSC (85.50% vs. SSFTT's 97.36%), and all evaluation is benchmark-based.

Future work: (1) activation distribution analysis to verify the β re-centering hypothesis; (2) CBAM/ECA-Net comparisons; (3) spatially disjoint evaluation on all datasets; (4) real agricultural UAV validation; (5) ≥ 30 seeds for reliable significance testing.

Supplementary information. All code, trained model checkpoints (6 datasets $\times$ 10 seeds = 60 models), machine-readable results, and high-resolution figures are publicly available at https://github.com/DevGuru-codes/research_paper_code.

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DECLARATIONS

- **Funding:** Not applicable.
- **Conflict of interest:** The authors declare no conflict of interest.
- **Ethics approval and consent to participate:** Not applicable.
- **Consent for publication:** Not applicable.
- **Data availability:** All benchmark HSI datasets used in this study are publicly available. Indian Pines and Salinas are available from Purdue University. Pavia University is available from the University of Pavia. Botswana and KSC are available from the GIC Hyperspectral Remote Sensing repository. WHU-Hi HanChuan is available from Wuhan University.
- **Materials availability:** Not applicable.
- **Code availability:** All source code is publicly available at https://github.com/DevGuru-codes/research_paper_code.
- **Author contribution:** Swati Hira contributed to the conceptualization, supervision, and review of this work. Devguru Tiwari contributed to the methodology, software implementation, experiments, data analysis, and writing of the manuscript. Both authors read and approved the final manuscript.

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Table XVIII. SMCNN performance: 5% vs. 15% training split

Dataset	OA (15%)	OA (5%)	Δ
Indian Pines	99.03%	96.17%	−2.86%
Pavia Univ.	99.89%	99.47%	−0.43%
Botswana	99.90%	98.59%	−1.32%
KSC	89.34%	85.50%	−3.84%
WHU-Hi	99.95%	99.56%	−0.39%

Table XIX. Transformer baselines: 5% vs. 15% training split

Dataset	Model	OA (15%)	OA (5%)	Δ
Indian Pines	SSFTT	98.57%	95.69%	−2.88%
Indian Pines	SpectralFormer	95.69%	89.13%	−6.56%
Pavia Univ.	SSFTT	99.80%	99.52%	−0.28%
Pavia Univ.	SpectralFormer	98.86%	95.50%	−3.36%
Botswana	SSFTT	99.59%	97.80%	−1.79%
Botswana	SpectralFormer	95.71%	82.13%	−13.58%
KSC	SSFTT	99.32%	97.36%	−1.96%
KSC	SpectralFormer	96.92%	84.48%	−12.44%

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